
BirdGEN: Conditional Spectrogram Augmentation for Bird Species Classification under Simulated Label Scarcity

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Abstract

BirdGEN investigates whether conditional spectrogram generation can improve bird-species classification under simulated label scarcity. We convert recording-isolated audio from three bird species into log-mel spectrograms and compare conditional VAE-v3 and Diffusion generators for classifier augmentation. Each newly initialized CRNN receives 50 real spectrograms per species, with or without additional generated samples. Across three real-data subset choices and three CRNN initialization seeds, validation selected 200 additional generated spectrograms per species for both generators. On a fixed 489-clip real test set, mean macro-F1 was 84.75% without augmentation, 87.48% with VAE-v3, and 86.21% with Diffusion. These results support classifier utility in the tested simulated low-resource setting, but do not establish performance on genuinely rare or unseen species.

Attestation of Teamwork

Shenghao Jin contributed to data preparation and developed and evaluated the conditional VAE. Harvey Li contributed to data preparation, the Residual CNN and CRNN classifiers, the low-resource augmentation experiment, and the VAE-Diffusion generator-only spectrogram-sampling comparison. Vincent Wang developed and evaluated the conditional Diffusion model. All members contributed to the preparation and revision of the final report.

1 Introduction and Problem Formulation

Passive acoustic monitoring can scale species detection beyond manual surveys. BirdNET demonstrates this potential with a deep neural network trained on extensive annotated bird-sound data to identify hundreds of species [1]. However, supervised systems depend on labeled examples, and ecological recordings are often imbalanced and expensive to label, leaving under-represented species with limited training data. Spectrogram augmentation offers one way to address this bottleneck. ECOGEN showed the potential of generated spectrograms for classification on real recordings, reporting a 12% average accuracy improvement [2].

Motivated by this downstream evaluation, BIRDGEN asks whether class-conditional VAE-v3 and Diffusion generators can improve classification when labeled data are scarce. To measure downstream classifier utility, we use the paired change in held-out macro-F1:

$$\Delta F1 = F1_{\text{real+generated}} - F1_{\text{real-only}},$$

where macro-F1 weights all three species equally.

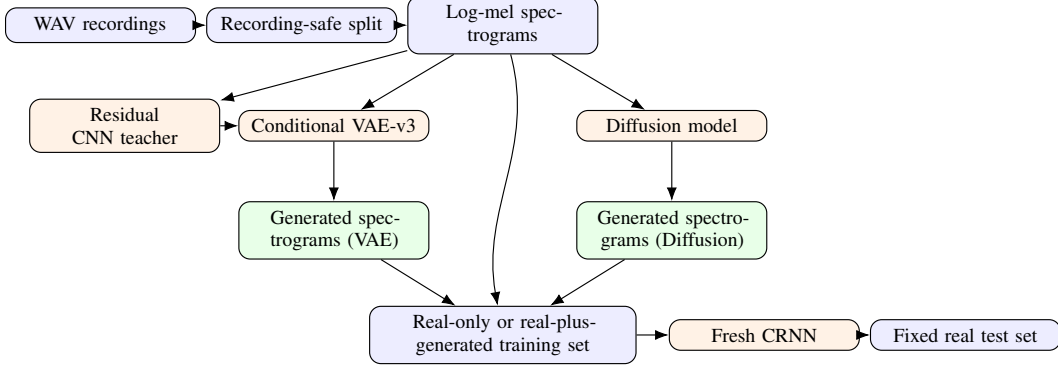


Figure 1: BirdGEN pipeline. Both generated-spectrogram branches feed the same downstream-classifier protocol.

2 Data Preparation

We use three-second, mono 22.05-kHz WAV clips from the Bird Song Data Set, compiled from Xeno-canto recordings [5]. We select American Robin, Northern Cardinal, and Song Sparrow, giving 3,347 clips from 310 source recordings. Splitting by recording ID prevents clips from the same recording from appearing across train, validation, and test sets.

Each clip is converted to a 128×128 log-mel spectrogram using a 1,024-point FFT and hop length 512. The VAE uses training-set global standardization, while the classifier maps each clip’s relative 80-dB range to $[-1, 1]$; an adapter converts VAE outputs to the classifier scale.

3 System Design and Models

Figure 1 shows the matched evaluation path. Every downstream condition trains a fresh CRNN and uses the same real test set.

3.1 Classifiers

The 1.66-M-parameter Residual CNN uses six residual blocks and global average/max pooling; its frozen features and logits guide VAE-v3 training. The selected 404-k-parameter CRNN combines convolutional spectral features with a bidirectional GRU and mean/max temporal pooling [6]. The selected CRNN is frozen for generator-quality diagnostics, whereas freshly initialized CRNNs provide the downstream augmentation evidence.

3.2 Conditional VAE Evolution

We selected a conditional VAE (CVAE) rather than an unconditional VAE, because every training spectrogram is paired with a bird-species label, and the downstream task requires controllable, class-labelled augmentation [3, 4]. The species label y conditions both the encoder $q_\phi(z \mid x, y)$ and decoder $p_\theta(x \mid z, y)$. Since the requested class is supplied explicitly, the latent variables do not need to represent species identity alone and can thus focus more on within-species variation, such as vocalization type, pitch, and timing. Conditioning reduces cross-species ambiguity, although it does not by itself guarantee realistic or diverse outputs.

V1 used a 128-dimensional global latent with concatenated class embeddings and an MSE+KL objective, but tended to oversmooth fine structure. V2 introduced a $16 \times 8 \times 8$ spatial latent, FiLM conditioning [7], detail-aware reconstruction losses, and a per-species Gaussian posterior sampler; however, a single Gaussian could still average distinct vocalization modes. These limitations motivated V3’s higher-resolution latent and posterior-anchor sampling.

V3 increased the latent resolution to $16 \times 16 \times 16$, used a lower KL weight with a longer warmup and per-channel free bits to reduce excessive latent compression and help retain information needed

for fine spectro-temporal detail. It also added feature and label consistency losses from a frozen, supervised Residual CNN species classifier trained on labeled real spectrograms, providing species-aware guidance. For generation, V3 replaced each single species-level Gaussian with a train-only posterior-anchor bank, allowing samples to be drawn near multiple encoded vocalization examples. This generation-time strategy mitigates the practical mismatch between the standard-normal prior and learned posteriors, but it does not replace the prior used to train the checkpoint. Since multiple components were modified between versions, the results show the improvement of each complete model rather than the individual causal effect of any single modification.

VAE-v3 has approximately 5.37 M parameters. Species embeddings modulate residual encoder and decoder blocks through FiLM. Its training objective is

$$\mathcal{L} = \mathcal{L}_{\text{event}} + 0.25\mathcal{L}_{\text{ms}} + 0.60(\mathcal{L}_{\Delta t} + \mathcal{L}_{\Delta f}) + \beta\mathcal{L}_{\text{KL,free}} + s_{\text{cls}}[0.08\mathcal{L}_{\text{feat}} + 0.03\mathcal{L}_{\text{label}}]. \quad (1)$$

Here, $\mathcal{L}_{\text{event}}$ is an event-weighted L1 reconstruction term, $\mathcal{L}_{\text{ms}}$ averages L1 losses at multiple resolutions, and $\mathcal{L}_{\Delta t}$ and $\mathcal{L}_{\Delta f}$ are edge-weighted L1 discrepancies between first-order time and frequency differences. The KL coefficient β is warmed up to 5×10^{-4} over 20 epochs, while s_{cls} increases to one over eight epochs. The free-bits term applies a 0.25-nat floor to the batch-and-spatial mean KL of each latent channel, so reducing a channel’s KL below this threshold provides no further optimization benefit. Multiscale and feature-space terms are motivated by [8, 9], and free bits follow [10].

For a training anchor i with the requested species label, the encoder defines a diagonal-Gaussian posterior $q_{\phi}(z | x_i, y) = \mathcal{N}(\mu_i, \text{diag}(\sigma_i^2))$. The corrected generation-time sampler applies temperature only to its stochastic residual:

$$z = \mu_i + T\sigma_i \odot \epsilon, \quad T = 0.35, \quad \epsilon \sim \mathcal{N}(0, I). \quad (2)$$

This correction affects post-training sampling only; the training-time reparameterization and VAE checkpoint were unchanged. The train-only, content-safe posterior bank retained 256 Northern Cardinal, 247 Song Sparrow, and 256 American Robin anchors, and the three generation-seed VAE pools were regenerated from this bank. The method is conceptually related to mixture-of-posteriors priors such as VampPrior [11], but uses fixed encoded training examples rather than learned pseudo-inputs and does not modify the training objective.

3.3 Diffusion Model

Architecture. The diffusion branch is a class-conditional DDPM [13] trained on the shared 128×128 standardized log-mel spectrograms. Its conditional U-Net [17] has base width 64, channel multipliers (1, 2, 4, 4), and two residual blocks per resolution, mapping $1 \times 128 \times 128$ inputs to a $256 \times 16 \times 16$ bottleneck and back through skip connections. Single-head self-attention at the bottleneck [18] captures global time-frequency structure. Combined sinusoidal timestep and learned species embeddings modulate every residual block through FiLM [7]. The model has approximately 18.4 million trainable parameters.

Training objective. A cosine schedule [15] adds Gaussian noise over $T = 1000$ steps:

$$x_t = \sqrt{\alpha_t} x_0 + \sqrt{1 - \alpha_t} \epsilon, \quad \epsilon \sim \mathcal{N}(0, I), \quad (3)$$

where $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$. For $t \sim \mathcal{U}\{1, \dots, T\}$, the network predicts the added noise from (x_t, t, y) by minimizing

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{x_0, y, t, \epsilon} [\|\epsilon - \epsilon_{\theta}(x_t, t, y)\|_2^2]. \quad (4)$$

Classifier-free guidance and sampling. During training, each species label is replaced by a null token with probability 0.1. Conditional and unconditional predictions are combined using classifier-free guidance [16]:

$$\hat{\epsilon} = \epsilon_{\theta}(x_t, t, \emptyset) + w[\epsilon_{\theta}(x_t, t, y) - \epsilon_{\theta}(x_t, t, \emptyset)]. \quad (5)$$

The EMA checkpoint (decay 0.999) supports both full 1000-step ancestral DDPM and deterministic 100-step DDIM sampling ($\eta = 0$) [14] without retraining, reducing reverse evaluations tenfold for DDIM. In both samplers, $\hat{x}_0$ is clipped to ± 4 standardized units to prevent extreme values and non-finite waveform inversions.

Table 1: Generator-quality diagnostics for the current three-seed generated pools.

Metric	VAE-v3	Diffusion
Selected-CRNN target-label accuracy / macro-F1	96.67% / 96.66%	92.44% / 92.42%
Classifier-feature Fréchet distance	22.8950	41.2550
Feature precision / feature recall	0.8528 / 0.8972	0.5654 / 0.8207
Feature density / coverage / diversity ratio	0.6423 / 0.7419 / 0.8719	0.3114 / 0.4845 / 1.0083
Pixel Wasserstein / exact copies	0.0787 / 0	0.1916 / 0

4 Experimental Methodology

For the primary low-resource experiment, each newly initialized CRNN received 50 real training spectrograms per species, selected from distinct recording IDs. We crossed three real-data subset choices with three CRNN initialization seeds, yielding nine matched settings. Validation compared real-only training with adding 50, 100, or 200 generated spectrograms per species. The CRNN architecture, masking policy, optimizer-step budget, and validation/test splits were held constant; no pretrained CRNN was augmented. The augmentation amount was selected using validation macro-F1, while the primary outcome was the paired change in macro-F1 on the fixed held-out test set. Only the shared real-only baseline and each generator’s selected condition were evaluated on the test set.

We also measured generator speed using five synchronized in-memory repeats of 600 spectrograms (200 per species) after warm-up, using FP32 precision, batch size 8, and the same RTX 4070 SUPER. Timing covered generation through the final normalized spectrogram tensor; loading, conversion, transfer, file output, waveform decoding, and metric computation were excluded.

5 Experimental Results

Classifier selection. The selected 404-k-parameter CRNN achieved a controlled three-seed validation macro-F1 of $90.45\% \pm 1.45\%$ and was used for the downstream augmentation experiments.

VAE diagnostics. Test reconstruction MSE in the globally standardized log-mel domain decreased from 0.1533 for V1 and 0.0654 for V2 to 0.0281 for V3, although the historical V1 evaluation sampled from its posterior and is not strictly matched. For a separate seed-42 diagnostic pool of 48 generated samples, mean same-species generated-to-test nearest-neighbor MSE was 0.5120, versus 0.3350 for the 489-query real-to-real leave-one-out reference. These metrics use different pairings, and nearest-neighbor pixel MSE can reward copying; neither directly measures perceptual or waveform realism. Appendix Figure 2 contrasts matched V1–V3 reconstructions and shows representative V3 generations.

Diffusion results. Development runs were compared using test noise-prediction MSE. The batch-8, 60-epoch run reduced this error from 0.41674 to 0.16647; the selected weighted-sampler, 100-epoch configuration had an MSE of 0.16667. With the same checkpoint, 100-step DDIM and 1000-step DDPM also had test noise MSE 0.16667. These values assess denoising error only and do not establish waveform realism or downstream augmentation utility.

Generator-quality diagnostics. The selected CRNN was used as an evaluator of generated spectrograms, not as the downstream test model. Table 1 summarizes class compatibility, feature-space distribution, and exact-copy checks for the current pools. Values are means over the three generation seeds; feature rows additionally average the three species.

Across the current generation seeds, Diffusion reached $92.44\% \pm 0.86\%$ target-label accuracy and $92.42\% \pm 0.84\%$ macro-F1, compared with $96.67\% \pm 1.20\%$ and $96.66\% \pm 1.21\%$ for VAE-v3. VAE-v3 had lower feature Fréchet distance, higher feature precision, recall, density, and coverage, and lower pixel Wasserstein distance; Diffusion had a diversity ratio closer to one (1.0083 versus 0.8719). These current-pool, classifier-view diagnostics describe different quality dimensions and do not establish waveform realism, perceptual quality, or downstream augmentation utility.

Table 2: Validation selection and held-out test results for low-resource CRNN augmentation.

Added generated spectrograms per species	VAE-v3 validation macro-F1	Diffusion validation macro-F1	VAE-v3 test macro-F1	Diffusion test macro-F1
0 (real only)	82.24%	82.24%	84.75%	84.75%
50	84.53%	85.45%	—	—
100	85.88%	86.06%	—	—
200 (selected for both)	86.47%	86.84%	87.48%	86.21%

Low-resource augmentation. Table 2 summarizes the validation selection and held-out test results. Each condition used 50 real spectrograms per species. Validation values are means across the nine matched experimental settings and determined the selected augmentation amount. The shared real-only baseline and each generator’s selected 200-generated condition were then evaluated on the fixed test set; dashes indicate conditions not evaluated on test. Diffusion had the higher validation macro-F1 at every nonzero augmentation amount, whereas VAE-v3 had the higher held-out test macro-F1. Mean held-out macro-F1 was 84.75% for real-only, 87.48% for VAE-v3, and 86.21% for Diffusion. The primary outcome, the paired change in held-out macro-F1, was $\Delta F1 = +2.72$ percentage points for VAE-v3 (7/9 positive blocks; descriptive block-bootstrap interval +1.00 to +4.65) and +1.46 points for Diffusion (6/9; +0.15 to +2.89).

Equal-count generator timing. The benchmark generated 600 fresh in-memory spectrograms (200 per species) per synchronized repeat after warm-up. Table 3 reports the complete request time, batch-average time per spectrogram, and throughput. The timing boundary ended at the final normalized spectrogram tensor and excluded model loading, conversion, transfer, file output, waveform decoding, and metric computation. Diffusion took $1348.09\times$ the VAE-v3 generator time for an equal-count request; measurements used FP32, batch size 8, five warm-up batches, five CUDA-synchronized repeats, and the same RTX 4070 SUPER.

Table 3: Generator-only timing comparison for matched fresh outputs.

Model	Mean time per 600 spectrograms (s)	Equivalent time per spectrogram (ms)	Throughput (spectrograms/s)
VAE-v3 (corrected sampler)	0.3058 ± 0.0096	0.510	1963.276
Diffusion (100-step DDIM)	412.3090 ± 2.8120	687.182	1.455

6 Discussion

In the simulated low-resource setting, both validation-selected 200-generated conditions improved mean held-out macro-F1 versus their matched real-only controls. The nine blocks reuse overlapping real subsets, so the intervals are descriptive and the largest tested ratio is not an estimated optimum. This result applies to newly initialized low-resource CRNNs, not the historical full-data CRNN. Reconstruction MSE rewards pixel proximity, the Residual CNN is coupled to V3 training and therefore restricted to its teacher role, and a closed-set independent CRNN cannot reject unknown or noisy inputs. Spectrogram and classifier metrics also do not replace listening or waveform-level evaluation [12]. VAE-v3 has higher CRNN target compatibility, lower macro-species Fréchet distance, and higher feature precision/coverage in these pools. Diffusion has a diversity ratio nearer one. Separately, Diffusion took $1348.09\times$ the VAE-v3 generator time under the recorded boundary.

7 Conclusion

BirdGEN now has corrected three-seed generator evidence, a completed matched low-resource CRNN study, and a matched timing comparison. Validation selected 200 additional generated spectrograms per species for both models; the mean paired macro-F1 changes were +2.72 points for VAE-v3 and +1.46 points for Diffusion. Quality diagnostics remain metric-specific rather than a composite winner, and the efficiency result is limited to generator-only spectrogram sampling on the recorded hardware.

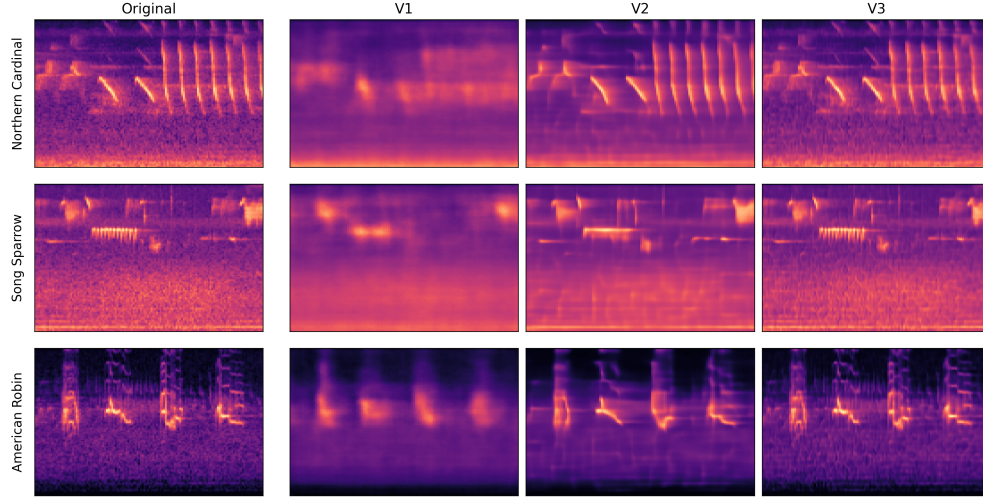
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A Supplementary VAE Diagnostics

VAE-v3 was trained with AdamW (learning rate 2×10^{-4} and batch size 24), with checkpoint selection based on the complete validation objective. The generated-to-test MSE used a separate, hash-recorded seed-42 notebook pool containing 16 samples per species and constructed from the train-only posterior bank; it is distinct from the three canonical augmentation pools.

(a) Matched held-out reconstructions



(b) VAE-v3 posterior-anchor generations

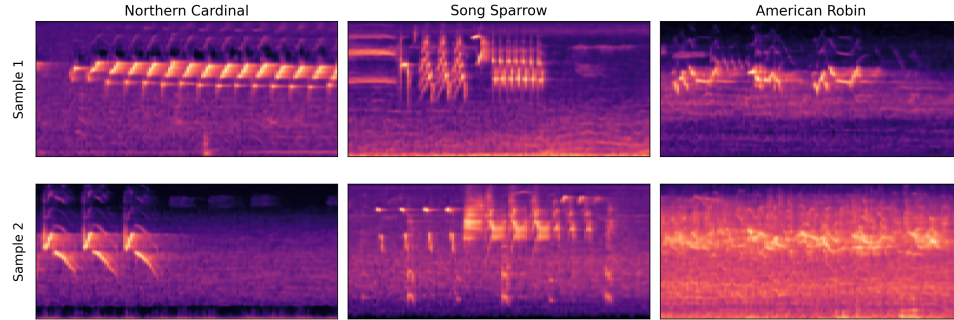


Figure 2: VAE evolution and conditional generation. (a) The same held-out example from each species is shown as the original and its V1–V3 reconstruction. Panels were re-rendered to a common display scale within each species row using the matched original as reference. V1 used posterior sampling, whereas V2 and V3 used posterior means, so this is a descriptive progression rather than a controlled ablation. (b) The first two displayed VAE-v3 posterior-anchor samples per species from the separate seed-42 diagnostic pool at temperature 0.35. Generated panels retain the notebook’s independent color normalization; appearance alone does not establish waveform realism.

B Diffusion Sampling Results

This appendix presents qualitative results for the DDIM-DDPM sampler comparison and the classifier-free guidance-weight comparison. Each sampling figure contains 12 representative generated spectrograms, with four examples per species. The trajectory figures show predicted clean spectrograms $\hat{x}_0$ at selected reverse schedule indices. The displayed columns are chosen snapshots and are not intended to represent every 100th reverse update. Their labels use the implementation’s zero-based schedule indices (0–999), corresponding to the one-based mathematical timesteps (1–1000) used in the equations. Because the initial noise samples were not fully paired across all configurations, these figures should be interpreted as qualitative examples rather than controlled sample-by-sample comparisons.

B.1 DDIM and DDPM Comparison

The DDIM results use the selected 100-step sampler. The DDPM results use the same trained checkpoint with only the reverse sampling procedure changed.

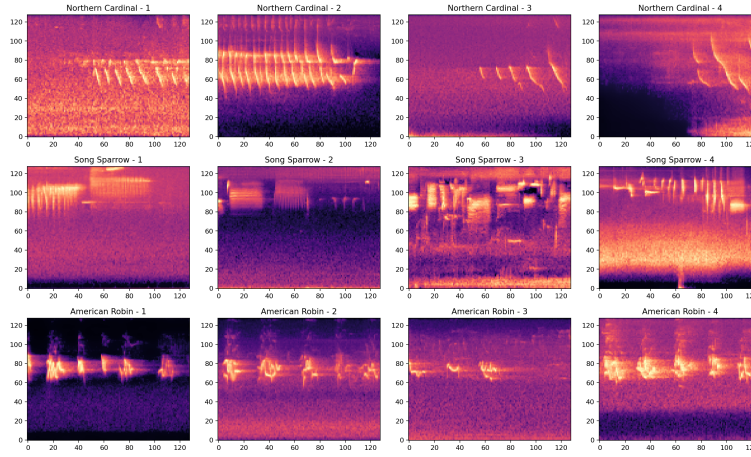


Figure 3: Representative spectrograms generated using 100-step deterministic DDIM sampling. Four examples are shown for each target species.

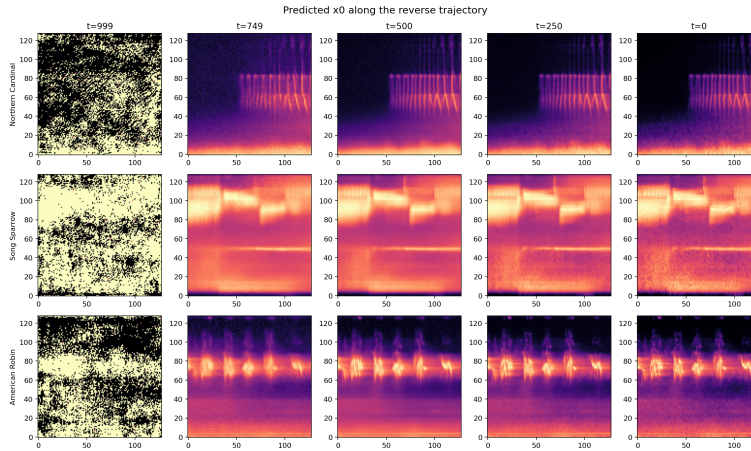


Figure 4: Predicted clean spectrograms $\hat{x}_0$ at selected timesteps along the 100-step DDIM reverse trajectory. Each row corresponds to one target species.

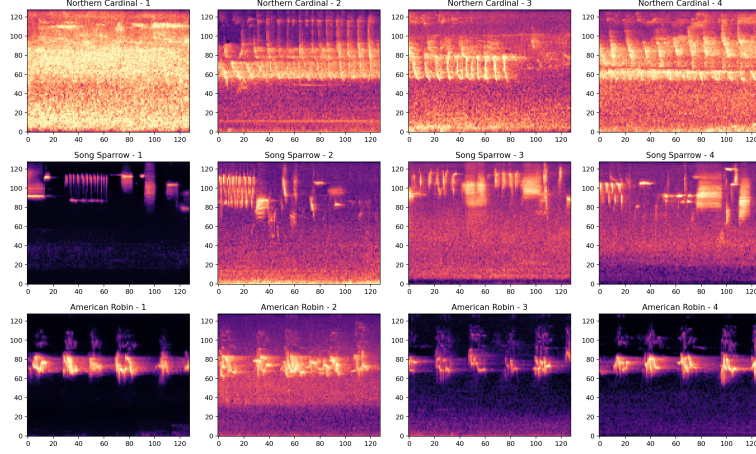


Figure 5: Representative spectrograms generated using full 1000-step ancestral DDPM sampling. Four examples are shown for each target species. The trained checkpoint is unchanged from the DDIM configuration; only the reverse sampling procedure differs.

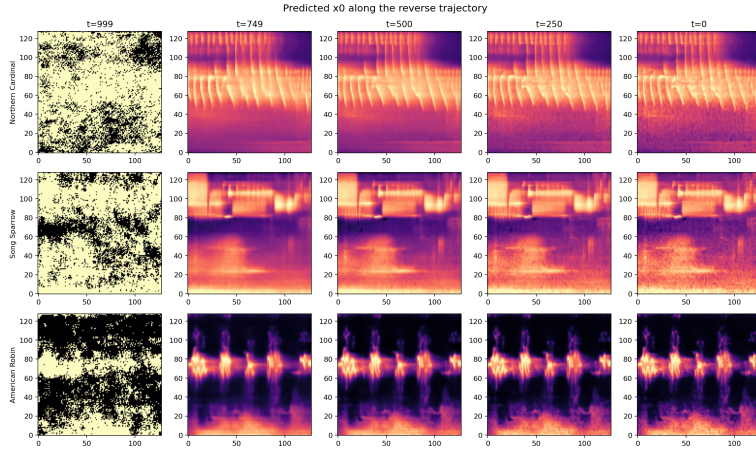


Figure 6: Predicted clean spectrograms $\hat{x}_0$ at selected timesteps along the full DDPM reverse trajectory. Each row corresponds to one target species. The trained checkpoint is unchanged from the DDIM configuration; only the reverse sampling procedure differs.

B.2 Classifier-Free Guidance-Weight Comparison

Figures 7–9 show representative outputs generated from the same trained checkpoint using 100-step DDIM sampling with classifier-free guidance weights $w = 0$, $w = 3$, and $w = 5$. The panels illustrate how guidance strength changes the generated spectrograms. For direct cross-reference, the $w = 3$ panel in Fig. 8 intentionally repeats the selected DDIM samples shown in Fig. 3; the two panels are therefore identical rather than independent sample sets.

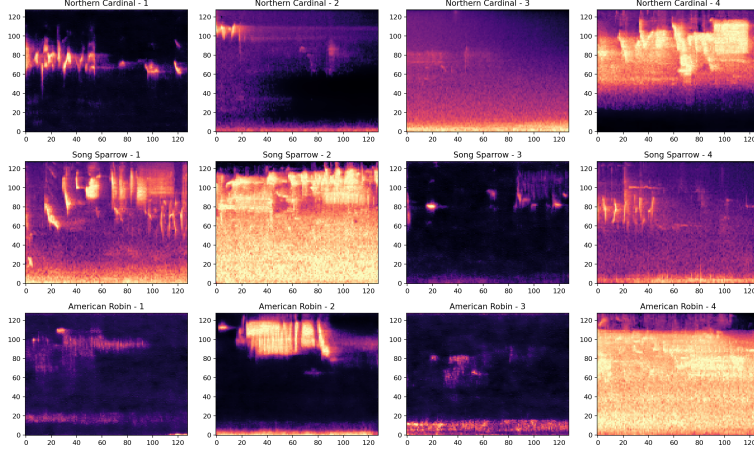


Figure 7: Representative spectrograms generated using 100-step DDIM sampling with classifier-free guidance weight $w = 0$. Under the formulation in Eq. (5), this setting uses the unconditional noise prediction and serves as a qualitative negative control for conditional generation.

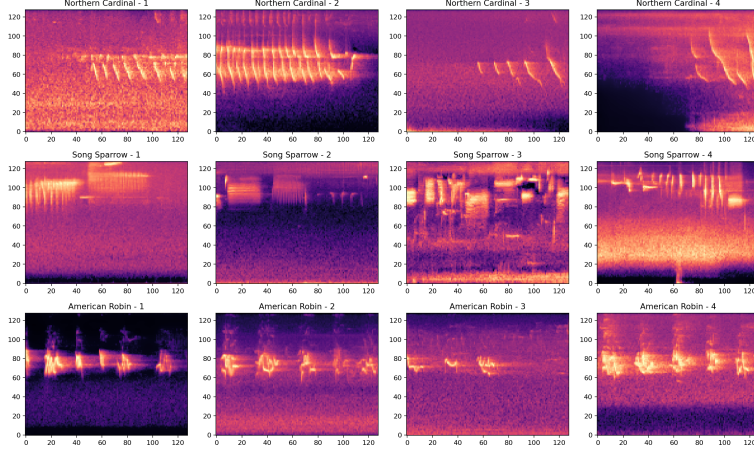


Figure 8: Representative spectrograms generated using 100-step DDIM sampling with classifier-free guidance weight $w = 3$. This setting was selected for the final generator. The panel intentionally repeats Fig. 3 for cross-reference.

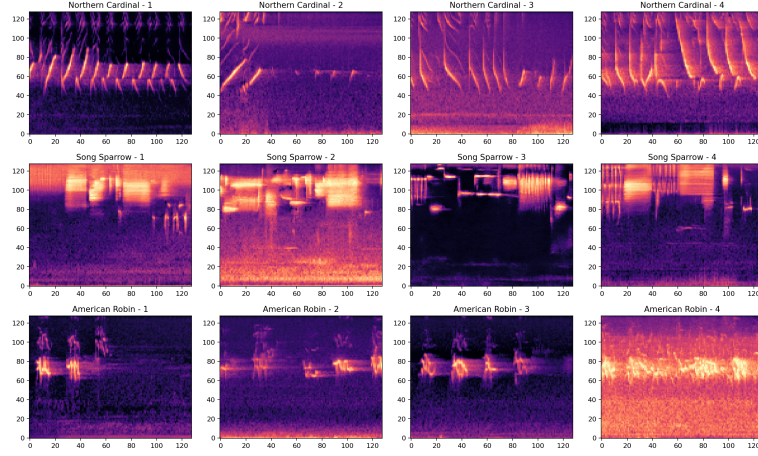


Figure 9: Representative spectrograms generated using 100-step DDIM sampling with classifier-free guidance weight $w = 5$. This setting uses stronger classifier-free guidance than the selected $w = 3$ configuration and provides a qualitative comparison of the generated spectrograms.